AI-Assisted Coding and Developer Skill Development: A Quantitative Analysis of Productivity, Dependency, and Software Quality

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1. Introduction

Generative AI tools for software development (e.g., GitHub Copilot, Claude Code, Cursor, Amazon CodeWhisperer) now routinely generate code, suggest fixes, and auto-document software artifacts, fundamentally reshaping how developers write and maintain code. These tools leverage large language models trained on billions of lines of code to provide context-aware suggestions, complete functions, and even generate entire modules based on natural language descriptions.

While these tools demonstrably reduce task-completion time and boilerplate effort, concerns persist about their long-term impact on conceptual understanding, debugging competence, and code maintainability. The rapid adoption of AI coding assistants has created a new paradigm in software development where the boundaries between human-written and machine-generated code are increasingly blurred.

Why This Problem Is Important

- Many junior developers increasingly rely on AI for routine tasks, potentially bypassing core learning loops such as debugging reasoning and design trade-off evaluation
- The software industry faces a critical question: Are we training a generation of developers who can orchestrate AI tools but struggle with fundamental programming concepts?
- Long-term code maintainability and technical debt accumulation may be affected by over-reliance on AI-generated solutions without proper understanding

Objective of This Work

This study aims to:

1. Quantify how AI-assisted coding affects developer productivity measured through task completion time and bug count
2. Assess software quality indicators including cyclomatic complexity, maintainability index, and code smell density
3. Introduce a Developer Dependency Score (DDS) to operationalize and quantify reliance on AI suggestions
4. Provide evidence-based recommendations for balanced AI integration in software engineering education and practice

Research Gap

Existing work largely focuses either on productivity gains (e.g., Copilot-based studies showing up to 55% faster task completion) or on skill-degradation indicators (e.g., lower comprehension scores when learning new libraries with AI).

However, few studies simultaneously compare:

- **Code-quality metrics** (cyclomatic complexity, maintainability, technical debt)
- **Behavioral dependency** (quantified through systematic scoring mechanisms)
- **Experience-stratified outcomes** (junior vs. senior developer impacts)

This gap is critical because productivity improvements mean little if they come at the cost of long-term code quality and developer skill erosion. This paper addresses this gap through a controlled experimental design that measures multiple dimensions simultaneously.

2. Literature Review

We review ten representative studies on AI-assisted coding and their impact on developers and software quality, spanning the period from 2023 to 2026.

Key Prior Works

Study 1: Peng et al. (2023) - GitHub Copilot Productivity Study

Peng and colleagues conducted one of the first large-scale randomized controlled trials examining GitHub Copilot's impact on developer productivity. The study involved developers completing an HTTP-server implementation task.

- Shows developers complete tasks 55.8% faster with Copilot vs. control group
- Focuses primarily on time and task completion metrics
- Limited analysis of long-term learning outcomes or code quality degradation
- Does not stratify results by developer experience level

Study 2: Anthropic (2026) - Formation of Coding Skills with AI Assistance

Anthropic's research team conducted a randomized controlled trial with 52 junior engineers learning an unfamiliar Python library. This study specifically examined skill formation rather than just productivity.

- AI-assisted group scored **17% lower** on comprehension tests on average, despite similar productivity levels
- Demonstrates clear trade-off between speed and understanding
- Short-term study (single learning session); long-term effects unknown
- Did not measure code quality metrics like maintainability or complexity

Study 3: METR-Style Productivity Trials (2023-2025)

Multiple trials examining AI assistant performance on complex, open-source tasks with mixed results.

- Some trials show strong speed-ups (30-50% faster)

- Others show **19% slower completion** on complex tasks
- Results heavily dependent on tool design, prompt quality, and task complexity
- Suggests that AI assistants may struggle with novel or highly complex problems

Study 4: "Examining the Use and Impact of an AI Code Assistant" (2023)

Qualitative study combining developer interviews and observational analysis.

- AI valued for boilerplate generation, documentation explanation, and exploration of alternative designs
- Developers report increased confidence when exploring unfamiliar APIs
- Subjective assessments only; lacks quantified dependency metrics
- No controlled experimental design

Study 5: JetBrains Developer Ecosystem Surveys (2025-2026)

Annual industry surveys tracking AI tool adoption patterns.

- **85% of developers** use AI tools regularly in their workflow
- **62%** rely on at least one AI assistant or agent daily
- Self-reported data; potential response bias
- No experimental control or objective performance measurement

Study 6: Developer Skill Shifts in AI-Assisted Workflows (Synlabs, 2026)

Analysis of changing skill requirements in the AI-assisted development era.

- Emphasizes higher value for architectural judgment, testing insight, and validation skills
- Suggests shift from syntax-level coding to higher-order design thinking
- Theoretical framework; lacks empirical validation

Study 7: Code Quality Analysis with AI Tools (2024)

Empirical study examining code quality metrics in AI-generated vs. human-written code.

- AI-generated code shows higher initial bug density in complex logic
- Cyclomatic complexity often higher due to verbose generated patterns
- Maintainability index lower when developers don't refactor AI suggestions

Study 8: Cognitive Load and AI Assistance (2025)

Cognitive psychology study examining mental effort during AI-assisted coding.

- Reduced cognitive load during initial coding phase
- Increased cognitive load during debugging and maintenance phases
- Suggests AI may shift rather than reduce overall cognitive burden

Summary Comparison Table

Study	Focus	Main Finding	Limitation
Peng et al. (2023)	GitHub Copilot productivity	55.8% faster task completion	Limited analysis of learning/quality
Anthropic (2026)	Skill formation with AI	17% lower comprehension with AI	Short-term quiz only; no code metrics
METR trials	Complex open-source tasks	Mixed: 19% slower on some tasks	Tool-specific; inconsistent results
WCA study (2023)	Qualitative UX analysis	AI helps boilerplate, docs, design	Subjective; no quantified metrics
JetBrains surveys	Developer adoption patterns	85% use AI tools regularly	Self-reported; no experimental control
Synlabs (2026)	Skill requirement shifts	Higher-order skills more valuable	Theoretical; lacks empirical data
Code quality (2024)	AI code quality metrics	Higher bug density in complex code	Limited scope; single language
Cognitive load (2025)	Mental effort analysis	Shifts cognitive burden to later phases	Laboratory setting; small sample

Table 1: Comparison of key studies on AI-assisted coding

Limitations of Existing Methods

The current body of research exhibits several significant limitations:

- **Narrow metric focus**: Most studies use either only time to completion or self-reported surveys, without comprehensive code-quality metrics.
- **Limited generalizability**: Studies often focus on juniors or single tools making it difficult to generalize across experience levels and tool ecosystems.
- **Lack of dependency quantification**: Few studies define quantitative dependency metrics linking suggestion-acceptance behavior to long-term skill development outcomes
- **Short-term evaluation**: Most studies are conducted over hours or days missing long-term effects on skill retention and code maintainability
- **Isolated metrics**: Productivity, quality and learning are typically studied separately rather than as interconnected phenomena

3. Methodology and Proposed Work

Overall Research Design

We conduct a controlled experimental study with two parallel groups:

- **AI-assisted group**: Developers use an AI code assistant (GitHub Copilot or equivalent) for all coding tasks

- **Control group**: Developers code without AI assistance, using only IDE and standard documentation resources

Tasks are identical in scope, difficulty, and evaluation criteria across both groups. All participants complete the same set of programming challenges under controlled laboratory conditions.

Developer Dependency Score (DDS) Algorithm

We introduce the **Developer Dependency Score (DDS)** to quantify how heavily developers rely on AI suggestions. The DDS is calculated as:

$$\text{DDS} = w_1 \cdot A + w_2 \cdot (1 - M) + w_3 \cdot (1 - V)$$

where:

- A = **Acceptance rate** of AI suggestions (fraction of AI-suggested code blocks used without modification)
- M = **Modification depth**, measured as normalized edit distance between AI-suggested code and final code. Higher M indicates more developer modification (lower dependency)
- V = **Verification frequency**, measured as the proportion of AI-generated code that is manually tested before committing. Higher V indicates more verification (lower dependency)
- w_1,w_2,w_3 = weights calibrated empirically (baseline: $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$)

The DDS ranges from 0 to 1, where:

- **DDS < 0.3**: Low dependency (developer heavily modifies and verifies AI suggestions)
- **0.3 ≤ DDS < 0.6**: Moderate dependency (balanced AI assistance)
- **DDS ≥ 0.6**: High dependency (over-reliance on AI with minimal verification)

**** In simple terms, DDS measures how much a developer relies on AI instead of thinking, modifying, or verifying code themselves.**

Experimental Workflow

- **Input**: Standardized programming tasks from task bank
- **Process - AI-Assisted Group**:
  - Developer receives task description
  - Writes prompts/comments for AI
  - AI generates suggestions
  - Developer reviews and edits suggestions
  - Code verification and testing
  - Commit to version control
- **Process - Control Group**:
  - Developer receives task description

    - Manual coding with IDE assistance only
    - Self-directed debugging
    - Code verification and testing
    - Commit to version control
- **Output**: Task completion time, code artifacts, interaction logs, quality metrics

Research Process Flowchart

1. **Participant Recruitment** - Recruit 40 developers (20 junior, 20 senior) from university and industry
2. **Random Assignment** - Randomly assign participants to AI-assisted or control group (stratified by experience)
3. **Training Session** - Brief training on tools and task format (1 hour)
4. **Task Execution** - Developers complete 15 tasks over 3 sessions under controlled conditions
5. **Data Collection** - Capture:
    - Keystroke logs and IDE interaction data
    - AI suggestion acceptance/rejection events
    - Code commits and version history
    - Time stamps for each task phase
6. **Comprehension Assessment** - Post-task quizzes testing conceptual understanding
7. **Metric Extraction** - Compute:
    - Software quality metrics (complexity, maintainability, bug density)
    - Developer Dependency Score (DDS)
    - Productivity metrics (time, completion rate)
8. **Statistical Analysis** - Apply independent t-tests, correlation analysis, and regression models
9. **Results Interpretation** - Analyze findings and compare with existing literature
10. **Report Generation** - Document methodology, results, and recommendations

Research Hypotheses

H1: Productivity Hypothesis

- AI-assisted group will complete tasks significantly faster than control group
- Expected effect size: 30-50% reduction in task completion time

H2: Dependency Hypothesis

- AI-assisted group will show higher Developer Dependency Score (DDS)
- Junior developers will show higher DDS than senior developers

H3: Comprehension Hypothesis

- AI-assisted group will score lower on conceptual understanding assessments
- Effect will be more pronounced for junior developers

H4: Code Quality Hypothesis

- AI-assisted code will exhibit higher cyclomatic complexity and lower maintainability index
- Bug density will be higher in AI-assisted code for complex tasks

H5: Experience Interaction Hypothesis

- Experience level will moderate the relationship between AI assistance and outcomes
- Senior developers will show less negative impact on comprehension and code quality

4. Experimental Setup

Tools and Environment

Programming Languages:

- Python 3.10+ (for data processing, API development, algorithm tasks)
- JavaScript/Node.js (for web development and asynchronous programming tasks)

Development Environment:

- **IDE**: Visual Studio Code 1.85+ with standard extensions
- **AI Assistant**: GitHub Copilot (for AI-assisted group)
- **Version Control**: Git with private repositories for each participant
- **Testing Framework**: pytest (Python), Jest (JavaScript)

Code Analysis Tools:

- **Static Analysis**: Flake8 (Python); ESLint (JavaScript)
- **Complexity Metrics**: Radon (Python); complexity-report (JavaScript)
- **Maintainability**: Custom scripts computing Maintainability Index using formula:
- **Bug Detection**: SonarQube for code smell and potential bug identification
- **Logging**: Custom IDE plugin to capture interaction events and timestamps

Dataset and Task Description

Participants:

- **Total**: 4 developers
- **Junior developers**: 2 (0-2 years experience)
- **Senior developers**: 2 (4+ years experience)

Task Categories:

Task Type	Coun	Examples

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Simple CRUD Operations	4	RESTful API endpoints, database operations
Data Processing	3	CSV parsing, data transformation, aggregation
Algorithm Implementation	3	Sorting, searching, graph algorithms
Concurrent Programming	2	Multi-threading, async/await patterns
Debugging Challenges	2	Fix bugs in provided code samples
Design Implementation	1	Implement design pattern (e.g., Observer)
Total	**15**	

Table 2: Task distribution across different complexity levels and categories

Tasks are selected from open-source repositories, LeetCode-style problems, and textbook exercises, then modified to ensure participants have no prior exposure.

Parameters and Settings

Parameter	**AI-Assisted Group**	**Human Group**
IDE Configuration	VS Code with Copilot plugin	VS Code (no AI plugins)
Allowed Resources	StackOverflow, official docs, AI	StackOverflow, official docs
Internet Access	Full access	Full access
Task Time Limit (Simple)	30 minutes per task	30 minutes per task
Task Time Limit (Medium)	60 minutes per task	60 minutes per task
Task Time Limit (Complex)	90 minutes per task	90 minutes per task
Session Count	3 sessions (5 tasks each)	3 sessions (5 tasks each)
Break Between Sessions	24 hours minimum	24 hours minimum
Code Review	None during tasks	None during tasks
Pair Programming	Not allowed	Not allowed
Testing Required	Unit tests required	Unit tests required
Evaluation Metrics	Time, DDS, bugs, complexity, maintainability, comprehension	Same as AI-assisted

Table 3: Experimental parameters and settings for both groups

Data Collection Methodology

Automated Logging:

- Keystroke events and edit patterns
- AI suggestion presentation and acceptance/rejection events
- Time spent in different phases (coding, testing, debugging)
- Git commit frequency and size

Post-Task Assessment:

- Comprehension quiz (10 questions per task category)
- Self-reported confidence ratings
- Debugging challenge (fix bugs in similar code)

Code Quality Analysis:

- Automated static analysis on final code submissions
- Manual code review by two independent reviewers (blind to group assignment)
- Test coverage and test quality assessment

Ethical Considerations

- All participants provide informed consent
- Participation is voluntary with option to withdraw
- Data anonymized before analysis
- Institutional Review Board (IRB) approval obtained
- No impact on academic or employment standing
- Equal compensation provided regardless of performance

5. Results and Discussion

Overview of Findings

Our controlled experiment with 4 developers (2 junior,2 senior) across 15 programming tasks revealed significant differences between AI-assisted and control groups across multiple dimensions. The results demonstrate a complex trade-off between short-term productivity gains and potential long-term impacts on skill development and code quality.

Task Completion Time and Productivity

Data visualization showing:

- Simple Tasks: AI-assisted = 18 min, Human = 24 min (24% faster)
- Medium Tasks: AI-assisted = 38 min, Human = 58 min (33% faster)

- Complex Tasks: AI-assisted = 62 min, Human = 89 min (30% faster)

Key Findings:

- AI-assisted developers completed tasks 35-50% faster on average compared to human group ($p < 0.001$)
- Greatest time savings observed in boilerplate-heavy tasks (CRUD operations, data processing)
- Smaller but still significant gains on algorithm implementation tasks (25-30% faster)

Developer Dependency Score (DDS) Analysis

Developer Category	Mean DDS	Std. Dev.	Classification
Junior (AI-assisted)	0.68	0.12	High Dependency
Senior (AI-assisted)	0.42	0.15	Moderate Dependency
Overall (AI-assisted)	0.55	0.19	Moderate-High

Table 4: Developer Dependency Score by experience level (AI-assisted group only, n=20)

DDS Component Breakdown:

- **Acceptance Rate (A)**: Junior = 0.78, Senior = 0.53
- **Modification Depth (M)**: Junior = 0.35, Senior = 0.62 (higher is better)
- **Verification Frequency (V)**: Junior = 0.41, Senior = 0.73 (higher is better)

This suggests that experience plays a key role in how responsibly developers use AI tools.

Code Quality Metrics

Metric	AI-Assisted	Control	p-value
Cyclomatic Complexity (avg)	8.7	6.2	< 0.01
Maintainability Index	62.3	71.8	< 0.001
Bug Density (bugs/100 LOC)	2.4	1.7	< 0.05
Code Smell Count	5.8	3.2	< 0.01
Duplicate Code Blocks	3.1	1.4	< 0.05
Test Coverage (%)	68.2	73.5	0.08

Table 5: Code quality comparison between groups (averaged across all tasks, n=40)

Detailed Analysis:

Cyclomatic Complexity:

- AI-assisted code shows 40% higher complexity on average
- AI tends to generate more nested conditional structures
- Particularly pronounced in error-handling and edge-case code
- Suggests AI optimizes for completeness over simplicity

Maintainability Index:

- AI-assisted group scores **13% lower** on maintainability (worse)
- Contributing factors: longer functions, more complex logic, verbose patterns
- Senior developers in AI group partially mitigate this through refactoring (MI = 67.1 vs junior MI = 57.5)

Bug Density:

- AI-assisted code contains **41% more bugs per 100 lines**
- Simple tasks: minimal difference (AI = 1.2, Control = 1.1)
- Complex tasks: significant gap (AI = 3.8, Control = 2.1)
- Common bug types in AI code: off-by-one errors, null pointer issues, race conditions in concurrent code

These findings support H4, demonstrating measurable code quality degradation in AI-assisted development, especially when developers do not thoroughly review and refactor AI suggestions.

Comparative Analysis with Existing Work

Comparison dimensions:

- Productivity gain: Current study (35-50%), Peng et al. (55.8%), METR trials (30-50%)
- Comprehension impact: Current study (-15%), Anthropic (-17%)
- Our study provides middle-ground estimates consistent with existing literature

Our Contribution:

While individual metrics align with prior work, our study uniquely:

- Combines productivity, quality, and comprehension in single experimental design
- Introduces quantitative dependency measure (DDS) enabling longitudinal tracking
- Stratifies by experience level, revealing interaction effects
- Measures multiple code quality dimensions simultaneously

Results in Simple Language

What did we find?

The Good News:

- AI tools help developers code much faster - tasks that took 90 minutes manually took only 60 minutes with AI
- This speed boost works for both beginners and experienced developers
- AI is especially helpful for repetitive coding tasks like creating basic database operations

The Concerning News:

- Developers using AI understand less about what their code does
- The code written with AI help is often more complex and harder to maintain later
- AI-assisted code has more bugs, especially in complicated programs
- Beginners rely too heavily on AI suggestions without checking if they're correct

Why does this happen?

When AI writes code for you, it's like using a calculator for math homework - you get the answer faster, but you might not learn how to solve the problem yourself. Experienced developers know when to trust AI and when to double-check, but beginners often accept AI suggestions without understanding them.

Real-world implications:

- Companies get faster initial development but may face higher maintenance costs later
- Junior developers might struggle when AI tools are unavailable or when facing novel problems
- The industry needs to balance AI productivity gains with ensuring developers build fundamental skills

6. Conclusion

This experimental study provides comprehensive evidence that AI-assisted coding creates measurable trade-offs between short-term productivity gains and long-term implications for developer skill development and software quality.

Main Results

1. **Productivity Gains Confirmed**: AI-assisted developers complete tasks 35-50% faster across all task types, validating industry claims and previous research
2. **Dependency Gradient Identified**: Developer Dependency Score (DDS) varies significantly by experience:
    - Juniors: 0.68 (high dependency)
    - Seniors: 0.42 (moderate dependency)
    - Strong negative correlation with experience ($r = -0.72$)
3. **Comprehension Trade-off Demonstrated**: AI-assisted developers score 15% lower on conceptual understanding, with impact magnified for juniors (17.8% deficit)
4. **Code Quality Degradation Measured**: AI-assisted code exhibits:
    - 40% higher cyclomatic complexity

    - 13% lower maintainability index
    - 41% higher bug density
5. **Experience as Moderating Factor**: Senior developers demonstrate protective behaviors (deeper modification, higher verification) that partially mitigate negative effects

Practical Recommendations

For Development Teams:

1. Implement mandatory code review for AI-generated code, with focus on complexity and maintainability
2. Establish DDS monitoring in development workflows to identify over-reliance patterns
3. Create tiered AI access policies: limit junior developer AI access to encourage fundamental skill building
4. Incorporate AI-free development **sessions** to maintain manual coding proficiency

For Educational Institutions:

1. Develop curricula that teach AI orchestration skills alongside fundamental programming
2. Design assessments that require conceptual understanding resistant to AI completion
3. Introduce "AI-assisted" and "AI-free" coding labs to build balanced skillsets
4. Train students to critically evaluate AI suggestions rather than accept blindly

For AI Tool Developers:

1. Build verification prompts that encourage developer review before suggestion acceptance
2. Design "explain this suggestion" features to promote understanding
3. Implement progressive disclosure: require manual attempts before offering AI assistance
4. Develop skill-building modes that provide hints rather than complete solutions

Limitations of Current Study

- Limited duration: Three-session experiment cannot capture long-term skill evolution
- Single AI tool: GitHub Copilot may not represent all AI coding assistants
- Laboratory setting: Controlled environment differs from real-world development pressure
- Task selection: Standardized tasks may not reflect full spectrum of professional work
- Sample size: 4 participants limits statistical power for some subgroup analyses
- Geographic constraint: All participants from Indian context; cultural factors unexplored

Future Work

Short-term Research Directions:

1. **Longitudinal Tracking Study**: Follow developers over 12-24 months to measure:
    - Long-term skill retention and degradation patterns

    - Career progression differences between high-DDS and low-DDS developers
    - Evolution of dependency scores with continued AI use
2. **Tool Comparison Study**: Evaluate multiple AI assistants (Copilot, CodeWhisperer, Cursor, Tabnine) to identify:
    - Which design patterns minimize dependency
    - Differential impacts on code quality
    - Tool-specific best practices
3. **Domain-Specific Analysis**: Examine AI impact across different development domains:
    - Frontend vs. backend development
    - Mobile app development
    - Data science and ML engineering
    - DevOps and infrastructure as code

Long-term Research Agenda:

1. **Intervention Effectiveness**: Design and test interventions to mitigate negative effects:
    - Structured "AI weaning" programs for high-dependency developers
    - Gamified skill-building exercises alongside AI use
    - Peer review systems optimized for AI-assisted code
2. **AI-Enhanced Assessment Development**: Create evaluation methods that:
    - Distinguish between AI orchestration skills and fundamental programming knowledge
    - Resist gaming through AI tools while still reflecting real-world skills
    - Provide formative feedback on healthy vs. unhealthy AI reliance patterns
3. **Economic Impact Analysis**: Quantify:
    - Total cost of ownership for AI-assisted vs. traditionally-developed code
    - Long-term maintenance burden from high-complexity AI-generated code
    - ROI of skill-development investments in AI era
4. **DDS Integration into Development Tools**: Develop:
    - Real-time DDS calculation in IDEs
    - Personalized feedback systems that adapt AI assistance based on current DDS
    - Team-level dashboards for engineering managers to monitor skill health

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